

NAIL FOLD CAPILLARY MICROSCOPY AND RAYNAUD PHENOMENON

▲ FIGURE 171-3 Nail fold capillary microscopy in a patient with scleroderma, showing capillary dropout with enlarged, dilated, tortuous capillary loops.

CLINICAL EVALUATION

In patients presenting with Raynaud phenomenon, evaluation should include assessment for **thoracic outlet syndrome**, which can cause vascular obstruction. The **Adson maneuver** is used to test for diminution of the radial pulse during exaggerated neck and shoulder movements.

A thorough **neurologic examination** is essential to identify: - Signs of sympathetic hyperactivity

- Abnormal reflexes
- Muscular weakness or atrophy
- Median nerve compression (e.g., carpal tunnel syndrome)

Skin examination may reveal clues to underlying systemic disease, such as: -

Telangiectases

- Skin texture changes (e.g., thickening or tightening)
- Rashes
- Purpura

These findings may suggest a **connective tissue disease** or **hyperviscosity syndrome**.

LABORATORY TESTS

All patients should undergo the following initial laboratory evaluation: -

Complete blood count (CBC)

- Erythrocyte sedimentation rate (ESR)

- **Urinalysis**
- **Anti-nuclear antibody (ANA) test**

Additional testing should be guided by clinical findings. In patients with a positive ANA, further testing for **specific nuclear antigens** is indicated, including:

- **Anti-Sm** (associated with systemic lupus erythematosus)
- **Anti-topoisomerase (anti-Scl-70)**
- **Anti-centromere antibodies** (associated with limited cutaneous scleroderma)

These help identify early **systemic lupus erythematosus (SLE)** or **scleroderma** (see Chaps. 156 and 158).

A **routine chest radiograph** should be obtained to evaluate for:

- **Cervical rib** (a cause of thoracic outlet syndrome)

- Signs of **interstitial lung disease**

NAIL FOLD CAPILLARY MICROSCOPY

This non-invasive test is valuable in distinguishing **primary** from **secondary Raynaud phenomenon**.

- In **primary Raynaud phenomenon**, capillary architecture is typically normal or shows only mild abnormalities.
- In **secondary Raynaud phenomenon** (especially due to connective tissue diseases), findings include:
 - **Enlarged, tortuous, dilated capillary loops**
 - **Capillary dropout (avascular areas)**
 - Progressive reduction in the number of capillary loops over time

Serial nail fold capillary microscopy can demonstrate progressive microvascular damage in secondary forms, particularly in **scleroderma** (Fig. 171-3).

Assessment of digital artery patency can be performed using **Doppler ultrasound**. **Digital subtraction arteriography** is reserved for patients with severe, prolonged ischemia being considered for arterial revascularization.

DIFFERENTIAL DIAGNOSIS

Raynaud phenomenon is classified as: - **Primary (idiopathic)**
- **Secondary** (associated with an underlying condition)

Primary Raynaud Phenomenon

Primary Raynaud phenomenon occurs in the absence of an identifiable underlying cause. Diagnostic criteria (combined from Allen and Brown¹⁶ and LeRoy and Medsger¹⁷) include:

TABLE 171-1: Criteria for Primary Raynaud Phenomenon

- Vasospastic attacks triggered by cold or emotional stress
- Bilateral involvement of extremities
- Normal vascular examination with symmetric peripheral pulses
- Normal nail fold capillary microscopy
- Absence of gangrene, or if present, limited to fingertip skin
- No evidence of underlying disease, drug use, or occupational exposure causing vasospasm
- Negative anti-nuclear antibody (ANA) test
- Normal erythrocyte sedimentation rate (ESR)
- History of symptoms for at least 2 years

Long-term studies show that approximately **15% of patients** with primary Raynaud phenomenon progress to a **secondary form** (most commonly a connective tissue disease like scleroderma) within the first decade.

Predictors of progression include: - Nail fold capillary abnormalities

- Hand swelling
- Positive Allen's test
- Positive ANA

Secondary Raynaud Phenomenon

CONNECTIVE TISSUE DISEASES

Connective tissue diseases are the **most common cause** of secondary Raynaud phenomenon (see Table 171-2).

- **Scleroderma (systemic sclerosis):**
 - 80–90% of patients have Raynaud phenomenon
 - It is the **presenting symptom in ~1/3 of cases**
 - May be the **only manifestation for years** before other features develop
- **Systemic lupus erythematosus (SLE), idiopathic inflammatory myopathies, and systemic vasculitis:**
 - Raynaud phenomenon occurs in **~30% of patients**

Early identification of secondary causes is critical for prognosis and management.